

Skills Summary

Converting Between Rectangular and Polar Coordinates

Use this reference while completing your worksheet or when studying.

The four conversion facts. $x = r \cos \theta$ $y = r \sin \theta$ $r^2 = x^2 + y^2$ $\tan \theta = \frac{y}{x}$

- Going **polar** $\rightarrow$ **rectangular** is pure substitution — there is nothing to decide.
- Going **rectangular** $\rightarrow$ **polar**, r is a distance and is never negative, but θ always needs a decision: two angles on $[0^\circ, 360^\circ)$ share the same tangent, half a turn apart.
- Sketch the point before writing θ . The quadrant is what picks between the two candidates, and no formula supplies it.

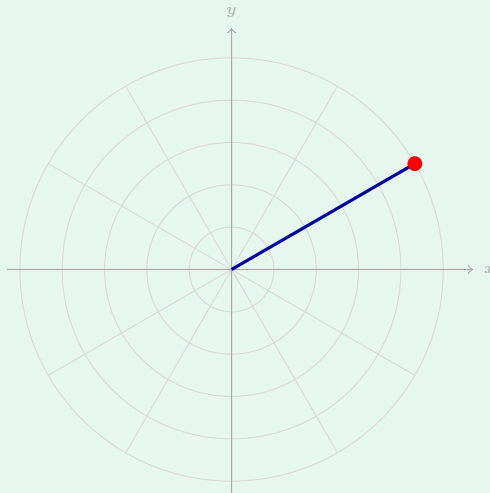
1. Polar to rectangular

Substitute into $x = r \cos \theta$ and $y = r \sin \theta$ and simplify. Keep exact values ($\frac{\sqrt{3}}{2}$, not 0.866) unless a decimal is asked for. Confirm with $x^2 + y^2 = r^2$, and check the signs against the quadrant of θ .

Example: Convert the polar point $(10, \frac{\pi}{6})$ to rectangular form.

Step 1. Nothing needs deciding here — the angle is given, so both coordinates come straight out of the two substitution formulas.

Step 2. $x = 10 \cos \frac{\pi}{6} = 10 \cdot \frac{\sqrt{3}}{2}$ and $y = 10 \sin \frac{\pi}{6} = 10 \cdot \frac{1}{2}$.



So the rectangular form is $(5\sqrt{3}, 5)$, in quadrant I as the picture shows.

Try it: Convert $(8, \frac{2\pi}{3})$ to rectangular form.

check: $(-4, 4\sqrt{3})$

2. Rectangular to polar

First $r = \sqrt{x^2 + y^2}$, always positive. Then take the reference angle $\arctan \left| \frac{y}{x} \right|$ and place it in the point's own quadrant: add nothing in QI, subtract from 180° in QII, add 180° in QIII, subtract from 360° in QIV.

Example: Convert the rectangular point $(-8, 6)$ to polar form, with the angle in degrees to two decimal places.

Step 1. The signs put the point in quadrant II, so the arctangent key cannot give the final angle — get the reference angle first and place it afterwards.

Step 2. $r = \sqrt{64 + 36} = 10$, and the key returns $\arctan \frac{6}{-8} = -36.87^\circ$; adding a half turn moves it into quadrant II. So the polar form is **$(10, 143.13^\circ)$** .

Try it: Convert $(-8, -6)$ to polar form, angle in degrees to two decimal places.

check: $(10, 218.67^\circ)$

3. Choosing the angle that fits the quadrant

$\tan$ repeats every 180° , so $\tan \theta = k$ always has *two* solutions on one turn, and they sit in diagonally opposite quadrants. Solve for both, then let the signs of x and y — or the sketch — keep one.

Example: A point has $\tan \theta = -\sqrt{3}$ and lies in quadrant IV. Find θ on $[0^\circ, 360^\circ)$.

Step 1. A negative tangent means x and y have opposite signs, which happens in two quadrants — so solve for both candidates before choosing.

Step 2. The reference angle for $\sqrt{3}$ is 60° , and the two solutions sit half a turn apart. The candidates are $\theta = 120^\circ$ or 300° , and only 300° lies in quadrant IV, so that is the polar angle.

Try it: $\tan \theta = 1$ and the point is in quadrant III. Give both candidates on $[0^\circ, 360^\circ)$, then pick the right one.

check: $\theta = 45^\circ$ or 225° ; quadrant III gives 225°

(!) Watch out: the arctangent key only ever answers between -90° and 90° , so it can never return a quadrant II or III angle — those always need a half turn added. And on the y -axis, where $x = 0$, $\frac{y}{x}$ does not exist at all: the angle is 90° or 270° , read straight off the picture.